Analítica de Big Data | FACEN-UNA | 2026 | Unidad 3

3.2 Procedimiento: Regresión Logística Paso a Paso

Pasos

Paso 1: Preparar y explorar

```
library(mlbench); library(caret); library(pROC)
data(PimaIndiansDiabetes)
str(PimaIndiansDiabetes)
table(PimaIndiansDiabetes$diabetes)  # distribución de clases
```

Paso 2: División y preprocesamiento

```
set.seed(42)
idx <- createDataPartition(PimaIndiansDiabetes$diabetes, p=0.8, list=FALSE)
train <- PimaIndiansDiabetes[idx,]
test  <- PimaIndiansDiabetes[-idx,]
```

Paso 3: Modelo base

```
m1 <- glm(diabetes ~ ., data=train, family=binomial)
summary(m1)
# Identificar variables significativas (p < 0.05)
```

Paso 4: Modelo reducido con variables significativas

```
m2 <- glm(diabetes ~ glucose + mass + age + pregnant,
          data=train, family=binomial)
# Comparar modelos con AIC (menor es mejor)
```

```
cat("AIC m1:", AIC(m1), "| AIC m2:", AIC(m2), "
")
# Odds Ratios del modelo reducido
round(exp(cbind(OR=coef(m2), confint(m2))), 3)
```

Paso 5: Predicción y evaluación

```
probs <- predict(m2, test, type="response")
pred <- factor(ifelse(probs>=0.5,"pos","neg"), levels=c("neg","pos"))
cm <- confusionMatrix(pred, test$diabetes, positive="pos")
print(cm)
```

Paso 6: Curva ROC y AUC

```
roc_obj <- roc(test$diabetes, probs)
cat("AUC:", round(auc(roc_obj),3), "
")
plot(roc_obj, col="#2E7D32", lwd=2)
# AUC > 0.8: bueno; > 0.9: excelente; 0.5: aleatorio
```

Paso 7: Ajustar umbral

```
for (u in c(0.3, 0.4, 0.5, 0.6)) {
  p <- factor(ifelse(probs>=u,"pos","neg"), levels=c("neg","pos"))
  cm_u <- confusionMatrix(p, test$diabetes, positive="pos")
  cat(sprintf("Umbral %.1f: Sens=%.3f, Spec=%.3f, Acc=%.3f
",
    u, cm_u$byClass["Sensitivity"], cm_u$byClass["Specificity"],
    cm_u$overall["Accuracy"]))
}
```